

Lab 6 — Perceptron using scikit-learn

Aim

To implement a **Perceptron classifier** using scikit-learn on a linearly separable binary classification dataset.

Algorithm / Procedure

1. Generate a binary classification dataset using `make_classification()`.
2. Split the dataset into training and testing sets.
3. Create a Perceptron classifier using scikit-learn.
4. Train the model using the training data.
5. Predict the classes of the test data.
6. Calculate the accuracy of the classifier.
7. Display the accuracy.

Program

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

# Generate dataset
X, y = make_classification(
    n_samples=200,
    n_features=2,
    n_redundant=0,
    n_clusters_per_class=1,
    random_state=1
)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)

# Create Perceptron model
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=1
)

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

OUTPUT

```
... Accuracy: 0.8833333333333333
```

Result

The scikit-learn Perceptron classifier was successfully trained and evaluated on a synthetic binary classification dataset, achieving high classification accuracy.